

IVAN HE

GPU SOFTWARE PERFORMANCE ENGINEER

GPU performance engineer working across graphics, games and machine-learning workloads — profiling, finding the bottleneck, and building the tooling that keeps it found.

San Francisco Bay Area
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EXPERIENCE

GPU SOFTWARE PERFORMANCE ENGINEER

AUG 2022 – PRESENT

Apple · Cupertino, CA · [Games, Graphics & Machine Learning org](#)

SOFTWARE SYSTEM DESIGNER

JUL 2021 – JUL 2022

AMD · Santa Clara, CA · [Data Center GPU Performance Engineering](#)

- Defined performance suites and profiled machine-learning and HPC workloads on AMD data-center GPUs to identify bottlenecks.
- Drove optimizations of ML and HPC models on AMD GPUs from those findings.
- Built tooling that streamlined collecting and assessing performance metrics across the team.

TECHNICAL MARKETING ENGINEER → TECHNICAL PRODUCT MANAGER

JUN 2019 – JUL 2021

KIOXIA America (formerly Toshiba Memory America) · San Jose, CA · [Client SSD](#)

- Bridged customer requirements and product development: reviewed product specifications, helped author MRDs and drafted technical responses to customer inquiries.
- Benchmarked client SSD performance and analysed test results against Toshiba's performance requirements.
- Produced technical marketing collateral that explained the advantages of the client SSD line.

CO-OP / INTERN ENGINEER

JUN 2017 – AUG 2017 · MAY 2018 – AUG 2018

AMD · Sunnyvale & Santa Clara, CA · [GPU performance & driver test](#)

- Built a model that flags anomalous GPU frame times and ranks statistically significant frames for draw-call trace analysis.
- Wrote Perl automation for GPU testing and benchmarking of upcoming game titles; tested and reviewed driver optimizations.
- Developed an auto-report tool on the .NET framework to speed up inter-departmental data flow.
- Built and debugged test systems from components.

EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY

2015 – 2019

B.A. Cognitive Science · Berkeley, CA

Certifications: Google Cloud Platform Fundamentals: Core Infrastructure · Architecting with Google Compute Engine (Specialization)

Skills: GPU performance analysis · C# / .NET · Java · Perl automation · Python · TypeScript / three.js

PROJECTS

TRIP PLANNER

[TYPESCRIPT](#) · [THREE.JS](#) · [OPENSTREETMAP](#) · [VERCEL FUNCTIONS](#) · [CLAUDE API](#)

A whole itinerary in one self-contained HTML file. github.com/JunxuanH/trip-planner

Day-by-day timelines paired with street maps drawn from real OpenStreetMap geometry, every booking link, hotel cards, budget and a prioritised booking list — plus a 20-slide deck that opens on a 3D map with the route drawn on. Serverless endpoints let the plan be edited on-site.

CHORDSMITH

[REACT](#) · [TYPESCRIPT](#) · [VITE](#)

Songwriting workspace: click a word, attach a chord. github.com/JunxuanH/chordsmith

Write lyrics, click any word to hang a chord above it, drop in guitar-tab blocks and transpose the whole song by semitone while keeping sharps, flats and slash-bass notation intact. Local-first, prints to a clean chart.

OKAYBUDDY

[NEXT.JS](#) · [EXPO](#) · [SUPABASE](#) · [LIVEKIT](#) · [CLAUDE](#) · [STRIPE](#)

Language exchange with voice rooms and in-chat corrections. · [private beta](#)

Web + mobile app on one Supabase backend: tap any message to translate it, suggest corrections on what your partner sends, and drop into live voice rooms hosted in your target language.

HEARTHLY

[NEXT.JS 15](#) · [REACT 19](#) · [SUPABASE](#) · [STRIPE](#) · [CLAUDE](#) · [RESEND](#)

Close, even when you're far apart. · [private beta](#)

A subscription web app for couples in long-distance relationships: a quiz-led onboarding funnel, an AI visit-plan generator, daily questions and “send a treat” — with Stripe trials and a warm, serif design system.